DISASTER MANAGEMENT

HYDROLOGICAL DISASTERS

MODULE - 6

DISASTER MANAGEMENT

DISASTERS ARISING FROM NATURAL HAZARDS

GEOPHYSICAL
- Earthquakes
- Volcano
- Tsunami

HYDROLOGICAL
- Avalanche
- Coastal Erosion
- Coastal Flood
- Hydrological flood
- Flash Flood
- Debris/mud flow

METEOROLOGICAL
- Cyclone
- Storm Surge
- Heat wave/ cold wave
- Heavy rains, Sand or dust storm

CLIMATOLOGICAL
- Droughts
- Extreme hot or cold conditions
- Forest fires
- Glacial lake outburst

BIOLOGICAL
- Epidemics – viral, bacterial, parasitic, fungal or prion infections
- Insect infestations
- Animal stampedes






RIVERINE FLOOD RISK

RIVERINE FLOOD RISK

1. Out of 40 million hectare of the flood prone area in the country, on an average, floods affect an area of around 7.5 million hectare per year.

2. Floods occur in almost all river basins of the country. Heavy rainfall, inadequate capacity of rivers to carry the high flood discharge, inadequate drainage to carry away the rainwater quickly to streams/rivers are the main causes of floods

3. The National Flood Control Programme was launched in 1954. Since then, sizeable progress has been made in the flood protection measures.

4. Climate change and the resultant increase in extreme weather events will naturally worsen the uncertainties associated with floods.

DISASTER MANAGEMENT

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THE DEVASTATING SERIES OF DELUGE IN INDIA, 2019

1. Floods wreaked havoc in most parts of the country in August 2019 after various rivers at 25 stations crossed their highest flood level (HFL), according to Central Water Commission (CWC) data.

2. Both Maharashtra and Karnataka suffered damages in devastating deluges including other states like Assam, Bihar, Kerala etc.

3. States like Rajasthan, Maharashtra, Karnataka, and Gujarat received 36, 30, 22, and 31 per cent more rainfall than normal between June 1 and September 18, 2019

4. Moreover, rise in average global temperatures led to a worrying trend of no rain for long periods and then a sudden bout of excessive rainfall causing extreme floods.

Floods are also a result of huge mismanagement of dams and lack of coordination between states.

BEST PRACTICES TO MITIGATE FLOODS

1. Ensure that flood-resistant features are incorporated in planning and execution of social housing schemes
2. Ensuring Multi-purpose Flood Shelters in which the state government will ensure proper maintenance
3. Proper alignment and design of waterways and drainage systems for roads, highways, and expressways
4. Undertake pre- and post-monsoon inspections of dams and reservoirs
5. Enhancing the safety of dams and reservoir
6. Desilting/ dredging of rivers to improve flow; drainage improvement; floodwater diversion through existing or new channels
7. Hazard resistant construction, strengthening, and retrofitting of all lifeline structures and critical infrastructure

DISASTER MANAGEMENT

FLOOD RISK REDUCTION

UNDERSTANDING DISASTER RISK

- Modernisation of flood forecasting and warning systems on a river basin basis
- Identification of priority flood protection and drainage improvement works
- Realtime collection of hydro-meteorological data on important rivers
- Preparation of large-scale hazard maps of flood prone areas identifying areas of high vulnerability.
- International cooperation to share warnings about rivers flowing from neighboring countries

INTERAGENCY COORDINATION

- Overall disaster governance by MOWR
- Organising and coordinating central assistance by MHA
- Effective dissemination of warnings, information and data by MOWR, IMD, DOS, MEITY, NDMA
- Coordination among central and state agencies for non structural measures - MHA, BIS
- NDMA to coordinate with SDMA and DDMA.
- Coordination among state agencies for ensuring updated norms/ codes and their implementation.

INVESTING IN DRR – STRUCTURAL MEASURES

- Flood control measures such as construction of embankments and levees
- Centre will provide technical support while the state agencies will be responsible for
- Short term - Identification suitable sites for temporary shelters for people and livestock evacuated from localities at risk
- Long Term - Construction of multi-purpose shelters in villages/ habitations prone to floods • Proper monitoring and maintenance of river embankments

DISASTER MANAGEMENT

FLOOD RISK REDUCTION

INVESTING IN DRR – NON STRUCTURAL METHODS

- Centre will be responsible for adoption of revised reservoir operation manuals
- Oversight and monitoring of compliance with coastal zone laws
- Promote institutional mechanisms for sharing forecasts, warnings, data, and information
- Regulatory framework for flood plain zoning and flood inundation management
- States will be responsible for Implementing land-use regulation for low lying areas as per flood control norms
- Review and modification of operation manuals for all major dams/ reservoirs .
- Promote private participation in disaster management facilities for afforestation measures.

CAPACITY DEVELOPMENT

- NIDM, LBSNAA, NDMA, NIC etc. will be involved in Training and orientation programs for central govt. staff
- SDMA, DDMA, Engineering Training Institutes, SIRD, Police Training Academies will be responsible for Training and orientation programs for state govt. staff, professionals for veterinary care and support to disaster affected animals
- NCC, NYKS, Scouts and Guides, and NSS erc. And other volunteers will be incorporated in search and rescue training programmes.
- Introduction of Crisis Management, emergency medical response/recovery and trauma management at Diploma /UG/ PG levels for Health Professionals and common awareness even at school level

URBAN FLOOD RISK

URBAN FLOOD RISK

1 Flooding in urban areas is caused by intense and/or prolonged rainfall, which overwhelms the capacity of the drainage system and hence affects large number of people due to high population density in urban areas.

2 Urban flooding is significantly different from rural flooding as urbanization leads to developed catchments, which increases the flood peaks and flood volumes. It is also seen that urban areas are receiving more flash floods.

3 Urban flooding is significantly upped by a combination of all factors like

- Meteorological Factors: Heavy rainfall, cyclonic storms
- Hydrological Factors: Overbank flow coastal city flood.
- Climate Change: due to various anthropogenic events

In 2010, the NDMA published separate guidelines for UFDM

BEST PRACTICES TO MITIGATE URBAN FLOODS

1. Implementing 'Mobile Walls' like in Germany, and 'sponge' cities in line with cities in China which involves replacing concrete pavements with porous pavements to ensure better filtration.
2. Restrict encroachments in natural drainage areas; clearance of river beds, proper implementation of Coastal Regulation Zone rules.
3. Artificial rain water harvesting by increasing storm drains, recharge trenches, retention basins, deepening of ponds and lakes
4. Afforestation practices - Planting drought resistant and flood resistant sturdy trees in vulnerable areas
5. Urban flooding will increase if wetlands are not protected. strong laws to protect urban lakes, their catchment and feeder channels will save the natural rain sponges - Wetlands

URBAN FLOOD RISK REDUCTION

UNDERSTANDING DISASTER RISK

- While Centre will provide technical support for mapping of possible inundation levels, the state agencies should develop land use planning based on multi-hazard disaster risk assessment
- Land-use planning maps to be placed in public domain.
- Ward Level Information System to be developed using high resolution satellite images integrated with socioeconomic data covering natural resources and infrastructure facilities
- The magnitudes of inundation levels due to various scenarios and causes will be simulated on GIS-based inundation model

INTERAGENCY COORDINATION

- Overall disaster governance and Organising and coordinating central assistance by MHUA
- SDMA, DDMA, Panchayats (peri-urban) for organising and coordinating the immediate response.
- MOES, MEITY, NDMA should be available for effective dissemination of warnings, information and data while the state and district machinery will deliver warnings to the last mile.
- All non-structural measures to be handled by MHUA, BIS, NDMA with the support of DMD, SDMA, DDMA, ULBs, Panchayats (periurban)

INVESTING IN DRR – STRUCTURAL MEASURES

- MHUA, BMTPC should be responsible for City Bridge Design Considerations, City Road Level Design and overall Technical Support .
- The state should handle all road re-levelling works or strengthening/ overlay works and
- Upgradation of the existing drainage and storm water systems.
- Establishing Emergency Operation Centres by NDMA and relevant ministries
- Ensure round the clock operations by SDMA, DDMA and Urban Local Bodies(ULBs)

Learning Space

URBAN FLOOD RISK REDUCTION

INVESTING IN DRR – NON STRUCTURAL METHODS

- MHUA will be responsible for preparation of comprehensive Urban Storm Drainage Design Manual (USDDM)
- Medium Term – designs considering current international practices, specific locations and rainfall etc.
- Long Term - MHUA, in consultation with States/UTs and ULBs promote storm water drainage information system based on best technologies available integrating spatial and non-spatial data
- Regular Environment Impact Assessment to be conducted by MOEFCC and storm water drainage concerns a part of all EIA norms
- Constituting an Urban Flooding Cell at State level and a DM Cell to be constituted at the ULB level

CAPACITY DEVELOPMENT

- Introduce UFDM modules in school curricula through CBSE
- ULB and Panchayats should involve in upgrading equipment and skills for UFDM
- Enlisting professionals for veterinary care and support to disaster-affected animals.
- SDMA and DDMA should carry out mass media campaigns to promote culture of disaster risk prevention.
- Strengthening network of civil society organizations for awareness generation about DRR and DM
- Ensure accurate documentation of all aspects of disaster events for creating good historical records for future research.
- Incorporating gender sensitive and equitable approaches in capacity development

Note: Coastal Erosion and Coastal Flood will be discussed in Module 8

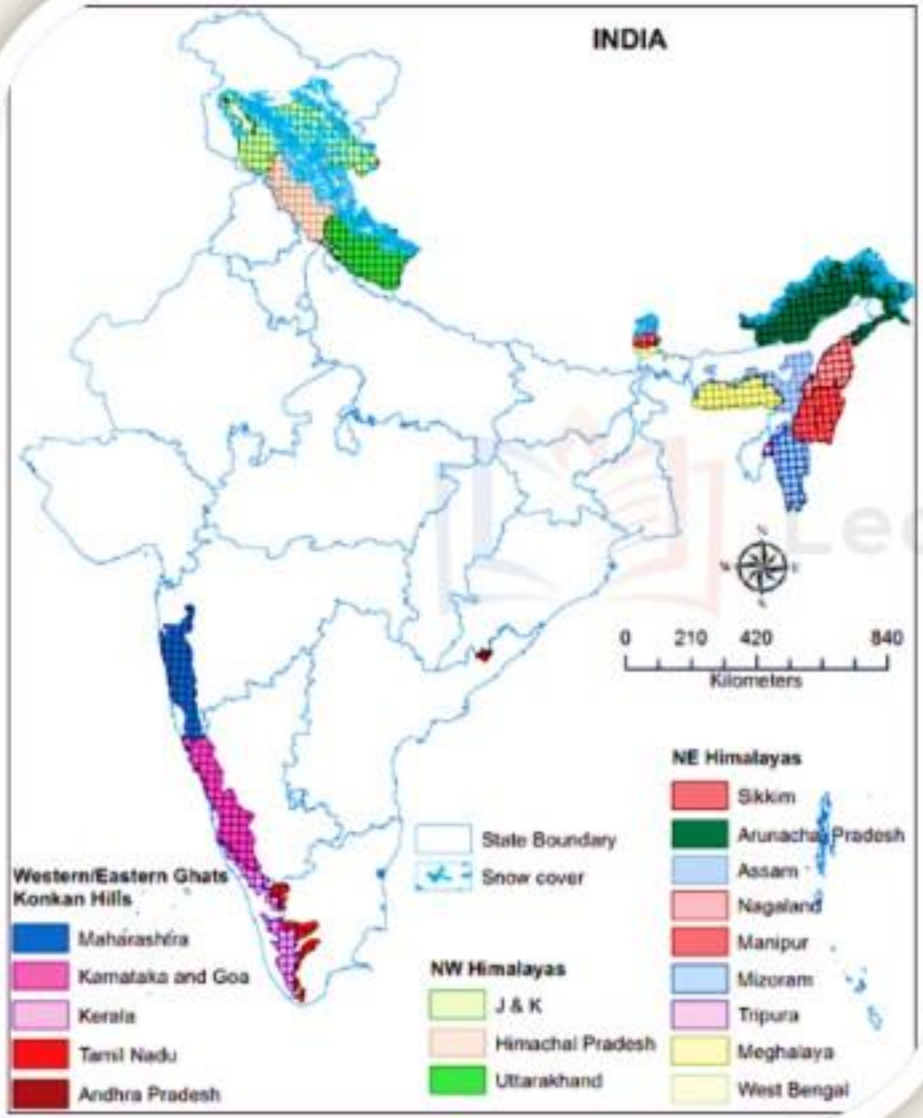

LANDSLIDES RISK

LANDSLIDES RISK

1. Landslides refer to the movement of mass of rock, debris or earth down the slope, when the shear stress exceeds the shear strength of the material.
2. Sheer stress is increased by removal of lateral and underlying support. Material Strength decreases due to the weathering, pore water pressure and changes in structure.
3. Landslides are increasing due to over urbanisation, massive deforestation, construction and development work in landslide prone areas.

MAJOR LANDSLIDE PRONE AREA OF INDIA

Learning Space



Source: NDMA

1. The Himalayas, North East India, Nilgiris and Western Ghats face the landslide disasters very often.
2. Due to climate change, unprecedented rains and plantations, removed vegetation are triggering landslides in Western Ghats and Nilgiris.
3. In India, excluding the permafrost regions in the north, about 0.42 Million km^2 areas of the landmass (12.6%) is landslide-prone

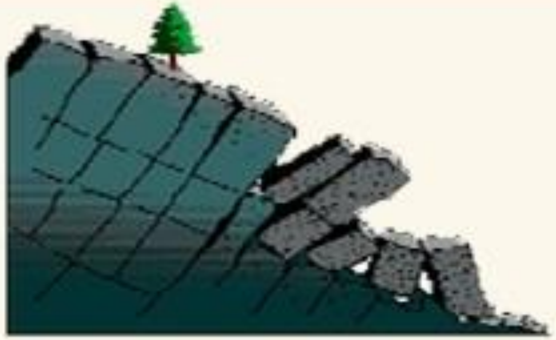
Learning Space

LANDSLIDES RISK

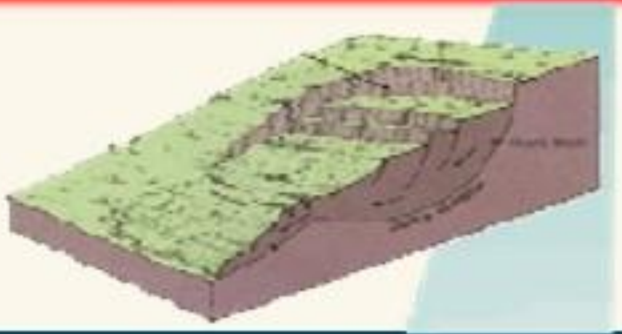
TYPES OF LANDSLIDES

1 Falls: abrupt movements of masses of geologic materials, such as rocks and boulders that become detached from steep slopes or cliffs.

2 Topples: forward rotation of a unit or units about some pivotal point, under the actions of gravity and forces exerted by adjacent units



3 Slides: rocks, debris or soil slide through slope forming material.

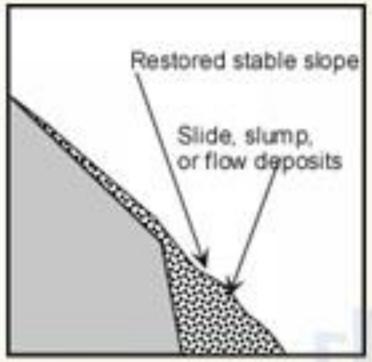


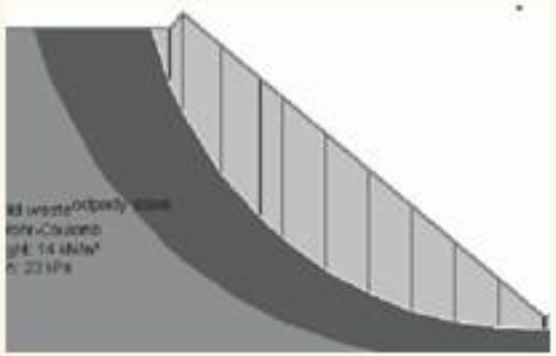
4 Spread: occur on very gentle slopes or flat terrain

DISASTER MANAGEMENT

LANDSLIDES RISK

REMEDIAL STEPS FOR LANDSLIDES





1 Modification of Slope Geometry: to improve the stability of the unstable or potentially unstable slopes, the profile of the slope is sometimes changed by excavation or by filling at the toe of the slope.

2 Retaining Walls: Construction of wall along the problematic slopes area.

3 Drainage Control: The presence of water in joints or in soil slope has a fundamental influence on the slope stability.

4 Internal Slope Reinforcement Systems: Rock slope stabilization with structural elements to help the rock mass to support itself by applying external structures which are not part of the rock mass


SNOW AVALANCHES RISK

SNOW AVALANCHES RISK

WHY THE RISK?

1. Avalanches are block of snow or ice descending from the mountain tops at a river like speedy flow.
2. They are extremely damaging and cause huge loss to life and property.
3. Beside killing people, avalanches also damage the roads, properties, and settlements falling in its way. Traffic blockage, structural damages of roads, and retaining wall damages occur most frequently due to avalanches.

HOW VAST THE RISK?

1. In Himalaya, avalanches are common in Drass, Pir Panijal, Lahaul-Spiti and Badrinath areas.
2. Western Himalaya – the snowy regions of Jammu and Kashmir, Himachal Pradesh and Uttarakhand, especially Tehri Garhwal and Chamoli districts
3. Jammu and Kashmir – Higher reaches of Kashmir and Gurez valleys, Kargil and Ladakh
4. Himachal Pradesh – Chamba, Kullu-Spiti and Kinnaur

DISASTER MANAGEMENT

LANDSLIDES AND SNOW AVALANCHE RISK REDUCTION

UNDERSTANDING DISASTER RISK

- Ministry of Mines and Ministry of Defence are the central agencies
- Unmanned Aerial Vehicle (UAV), Terrestrial Laser Scanner, and very high resolution Earth Observation (EO) data to be used.
- Landslide Hazard Zonation maps to be prepared at macro scale (1:50,000 / 25,000) ,and at meso level (1:10,000)
- Promote studies, documentation and research for Hazard Risk Vulnerability and Capacity Assessment (HRVCA)
- Creation of Special Purpose Vehicle (SPV) for Landslide Management

INTERAGENCY COORDINATION

- MoM and MoD is involved in the overall governance, coordination, technical inputs, and support.
- Organising and coordinating central assistance will be provided by MHA.
- DMD, SDMA, RD, DDMA, Panchayats, ULBs will be involved in Preparation and implementation of DM plans
- Geological Survey of India, Border Roads Organisation, IMD along with NDMA should be involved in warnings, information and data, Non-structural measures etc.

INVESTING IN DRR – STRUCTURAL MEASURES

- Central agencies should look after Protection of Human Settlements while state should be involved in Improving infrastructure, roads, and land stabilization work
- Archeological Survey of India should be responsible for Protection of Heritage Structures
- NDMA and NIDM will provide technical support for multi-hazard shelters where state agencies will identify safe buildings and sites
- SDMA, DDMA, Panchayats,ULBs will be responsible for proper maintenance of roads

LANDSLIDES AND SNOW AVALANCHE RISK REDUCTION

INVESTING IN DRR – NON STRUCTURAL METHODS

- MoM and MoD will plan site selection for Human Settlements in Landslide and Snow Avalanche Prone Areas.
- SDMA and DDMA will adopt suitable byelaws for rural and urban areas and enforce model codes into practice
- Codes and guidelines published by BIS will revise/revalidate every five years or earlier, if necessary
- Central agencies will establish a Professional Civil Engineers Council established by an Act for certification of engineers and evolve a procedure for certification of engineers.
- Include Public- Private partnerships to ascertain faster completion of projects.

CAPACITY DEVELOPMENT

- Train professionals on how to handle slope failures and their remediation and landslide emergencies by promoting observational method of design and construction with training on the development of contingency plans
- Include information on landslides and snow avalanches in the curriculum
- Promote use of insurance/ risk transfer and promote Community Radio for mass awareness.
- Promoting the planning and execution of emergency drills by all ministries and in all States/UTs
- State agencies should carry out post landslides field investigations, document the lessons learnt and disseminate.

RELATED QUESTIONS THAT CAN BE ANSWERED AFTER READING THE MODULE

1. Describe structural and non-structural mitigation measures in disaster management (of any given disaster)
2. Disaster preparedness is the first step in any disaster management process. Explain how hazard zonation mapping will help disaster mitigation in the case of landslides.(2019)
3. With reference to National Disaster Management Authority (NDMA) guidelines, discuss the measures to be adopted to mitigate the impact of recent incidents of cloudbursts in many places of Uttarakhand. (2016)
4. Urban Floods are getting more frequent and more devastating in India. What are measures that are taken to address urban flooding. Elucidate the NDMP guidelines to mitigate urban flooding.

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For Suggestions:
suggestions@learningspace.in

To Contact us:
info@learningspace.in

For Information:
info.learningspacedigital@gmail.com

Visit us at:
www.learningspacedigital.com
0866 244 44 72
98499 42299
94415 27588